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Non-body-fitted Cartesian-mesh simulation of highly turbulent flows using multi-relaxation-time lattice Boltzmann method

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ABSTRACT

This paper presents a lattice Boltzmann method (LBM) based study aimed at numerical simulation of highly turbulent and largely inclined flow around obstacles of curved geometry using non-body-fitted Cartesian meshes. The approach features (1) combining the interpolated bounce-back scheme with the LBM of multi-relaxation-time (MRT) type to enable the use of simple Cartesian mesh for the flow cases even with complex geometries; and (2) incorporating the Spalart–Allmaras (SA) turbulence model into LBM in order to represent the turbulent flow effect. The numerical experiments are performed corresponding to flows around an NACA0012 airfoil at $\mathrm{Re} = 5 \times 10^5$ and around a flat plate at $\mathrm{Re} = 2 \times 10^4$, respectively. The agreement between all simulation results obtained from this study and the data provided by other literature demonstrates the reliability of the enhanced LBM proposed in this paper for simulating, simply on Cartesian meshes, complex flows that may involve bodies of curved boundary, high Reynolds number, and large angle of attack.

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1. Introduction

The lattice Boltzmann method (LBM) is a mesoscopic approach based on kinetic theory of fluids. In LBM, fluid is considered as a group of discrete particles, and the flow domain is discretized as a number of lattices. By simulating the collision and advection processes across a finite number of discrete particles with the aid of particle distribution functions, the macroscopic density, velocity, and other physical quantities can be obtained. As a more intuitive approach, LBM also features the readiness in handling the interaction between fluid and other objects that are involved in a fluid flow scenario. In spite of as short a history as about 20 years, LBM has been successfully applied in many areas of fluid mechanics owing to its nature of parallelism, simplicity of programming and boundary treatment, as well as its capability of being incorporated with other fluid flow models aimed at extensive functionalities.

The efforts in improving LBM have been unintermittent since its inception. The single-relaxation-time (SRT) scheme based on LBM has been routinely used; however, due to its insufficient numerical stability, this method seems unlikely to satisfactorily simulate flows of more challenging nature, such as highly turbulent flow. In 2000, Lallemand and Luo [1] found that the multi-relaxation-time (MRT) model proposed by d'Humeriers [2] is more suitable than SRT in terms of physical principles, parameter selection, and numerical stability. Thereafter, MRT has been exhibiting an increasing popularity especially in LBM simulations of high Reynolds number flows.

In LBM applications, boundary treatment also plays a crucial role as it directly impacts the numerical stability and, hence, the solution accuracy. For the class of curved boundaries on Cartesian mesh, Filippova et al. [3] proposed an interpolation

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method based on the virtual equilibrium distribution. Then, this strategy was improved by Mei et al. [4] with a particular relation established between the relaxation factor of the interpolation and the dimensionless relaxation time in LBM. Guo et al. [5] combined the non-equilibrium extrapolation method with the interpolation of macroscopic quantities for curved boundary treatment. Later, Bouzidi et al. [6] proposed a new approach that interpolates the distribution function in the direction of lattice vector penetrating the wall, and this method is also referred to as interpolated bounce-back (IBB) scheme. On the other hand, Lallemand and Luo [7] suggested using the distance of a fluid node to the boundary to classify the interpolation into two categories: before collision and after collision. This method features second-order numerical accuracy, and can also be used to solve the moving boundary problems; however, this scheme may lead to jump of distribution function, potentially resulting in unphysical oscillation. In order to overcome this drawback, Yu et al. [8] assumed that, after a collision, the distribution function coming from the fluid side may cross the wall of solid body and then reach the node inside the solid body, which builds up a unified boundary treatment scheme; this method is sometimes also referred to as YMS scheme.

These boundary treatment approaches associated with LBM enables a Cartesian mesh to perform simulations of fluid flows around obstacles of irregular geometry; however, as of today, fairly little application using such a unified boundary treatment scheme has been found in simulating turbulent flows. In reality, turbulence is encountered in a large number of fluid mechanics applications. In numerical simulations, the computational cost of direct numerical simulation (DNS) of turbulent flow is apparently high. Alternatively, most of turbulent flow simulations via LBM were carried out with the aid of a variety of turbulence models or large eddy simulation (LES). Teixeira [9] successfully simulated the turbulent flow in a straight pipe by incorporating the two-layer mixing-length algebraic model and the two-equation k-ε model into LBM. This strategy was then extended by Filippova et al. [10], leading to multi-scale lattice Boltzmann schemes which combine the k-ε turbulence model and the virtual equilibrium scheme; as a result, [10] carried out numerical investigation of turbulent flow passing over some complex curvilinear geometries. Imamura et al. [11] presented the generalized form of interpolation supplemented LBM and combined it with the Baldwin–Lomax (BL) turbulence model to simulate the turbulent flow around an NACA0012 airfoil. Also, Shu et al. [12] combined the k-ε and Spalart–Allmaras (SA) turbulence models with the Taylor series expansion and least-squares-based lattice Boltzmann method for simulations of turbulent flows. On the other hand, Yu et al. [13] used the MRT model and LES to simulate a square jet, while Weickert et al. [14] incorporated LES along with the wall-adapting local eddy-viscosity (WALE) turbulence model into LBM. In addition, a model called very large eddy simulation (VLES) was presented in conjunction with LBM, as reported in [15–19]. Recently, Zhuo et al. [20] have incorporated the SA turbulence model into LBM to simulate high Reynolds number compressible flows around an NACA0012 airfoil using the body-fitted mesh.

This study focuses on LBM simulation of turbulent flows at high Reynolds numbers with the aid of the SA one-equation turbulent flow model as well. Unlike the aforementioned LBM applications, the present LBM for turbulent flow simulation adopted the unified boundary treatment scheme, which no longer relies on body-fitted meshes and, thus, renders LBM more flexible especially in handling the cases with complex geometries and/or challenging flow conditions.

The SA turbulence model has been successfully employed in [21]. Not only does the SA model feature the simplicity in computing the eddy kinematic viscosity, but also the capability of locally dealing with one point without involving any other points on the computational domain. Therefore, this turbulence model exhibits great flexibility in the choice of mesh as well as remarkable reduction in the cost of computation. Furthermore, it was reported in [22] that the SA model performs well for simulations of a big variety of flows, and can even outperform some two-equation turbulence models in predictions of flow separation and re-attachment. For such reasons, the SA model was selected to be incorporated into LBM for simulating highly turbulent flows in this study.

The rest of the paper is organized as follows. The MRT LBM model along with the YMS scheme and the SA turbulence model are described in the Section 2. Section 3 presents selected numerical simulations corresponding to flows with large Reynolds number and at large incidence; the simulation results are compared against data reported in other studies [23,24] to demonstrate the usefulness and effectiveness of this present work. Finally, some remarks concluded from this study are grouped in Section 4.

2. Numerical methods

2.1. LBM model

For the two-dimensional nine-velocity (D2Q9) model (see Fig. 1), the discrete velocities are defined as

$$[\boldsymbol{e}_0, \boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3, \boldsymbol{e}_4, \boldsymbol{e}_5, \boldsymbol{e}_6, \boldsymbol{e}_7, \boldsymbol{e}_8] = c \begin{bmatrix} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{bmatrix} \tag{1}$$

where the lattice speed c is related to the sound speed c_s using $c_s = c/\sqrt{3}$. The equilibrium distribution function can be obtained by

$$f_i^{(\mathrm{eq})}(\boldsymbol{x}, t) = w_i \rho \left[1 + \frac{\boldsymbol{e}_i \cdot \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{e}_i \cdot \boldsymbol{u})^2}{2c_s^4} - \frac{u^2}{2c_s^2} \right], \quad i = 0, 1, \ldots, 8, \tag{2}$$

with $w_0 = 4/9$, $w_1 = w_2 = w_3 = w_4 = 1/9$, and $w_5 = w_6 = w_7 = w_8 = 1/36$.

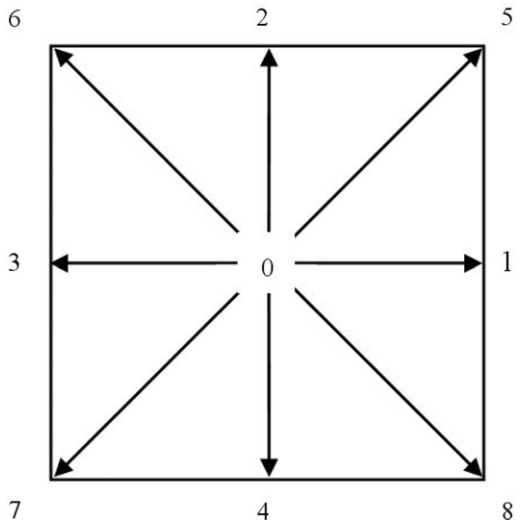


Fig. 1. D2Q9 lattice model.

In the MRT model, the distribution function space $|f\rangle$ is denoted by

$$|f\rangle = (f_0(x,t), f_1(x,t), \ldots, f_8(x,t))^T. \tag{3}$$

A moment space $|m\rangle$ is then defined [25] as

$$|m\rangle = (\rho, e, \varepsilon, j_x, q_x, j_y, q_y, p_{xx}, p_{xy})^T, \tag{4}$$

where ρ is the density, e is the energy, ε is the square of the energy, j_x, j_y and $q_x,\ q_y$ correspond respectively to the components of momentum and energy flux in the x- and y-directions, while p_{xx} and p_{xy} correspond to the diagonal and off-diagonal components of the viscous stress tensor. The moment space is related to the distribution function by the following 9×9 transformation matrix:

$$\boldsymbol{M} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -4 & -1 & -1 & -1 & -1 & 2 & 2 & 2 & 2 \\ 4 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & -2 & 0 & 2 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \\ 0 & 0 & -2 & 0 & 2 & 1 & 1 & -1 & -1 \\ 0 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \end{pmatrix}, \tag{5}$$

satisfying:

$$|m\rangle = \boldsymbol{M}|f\rangle. \tag{6}$$

The evolution equation of the MRT model reads:

$$|f(\boldsymbol{x} + \boldsymbol{e}\delta t, t + \delta t)\rangle - |f(\boldsymbol{x}, t)\rangle = -\boldsymbol{M}^{-1}\boldsymbol{S}[|m(\boldsymbol{x}, t)\rangle - |m^{(\mathrm{eq})}(\boldsymbol{x}, t)\rangle], \tag{7}$$

where $\boldsymbol{S} = \mathrm{diag}(s_0, s_1, \ldots, s_8)$ is a diagonal relaxation matrix. Let $|m^{(\mathrm{eq})}\rangle$ represent the equilibrium values of $|m\rangle$, which can be written as

$$|m^{(\mathrm{eq})}\rangle = (\rho, e^{(\mathrm{eq})}, \varepsilon^{(\mathrm{eq})}, j_x, q_x^{(\mathrm{eq})}, j_y, q_y^{(\mathrm{eq})}, p_{xx}^{(\mathrm{eq})}, p_{xy}^{(\mathrm{eq})})^T, \tag{8}$$

with

$$\begin{aligned} e^{(\mathrm{eq})} &= -2\rho + 3(j_x^2 + j_y^2) \\ \varepsilon^{(\mathrm{eq})} &= \rho - 3(j_x^2 + j_y^2) \\ q_x^{(\mathrm{eq})} &= -j_x \\ q_y^{(\mathrm{eq})} &= -j_y \\ p_{xx}^{(\mathrm{eq})} &= j_x^2 - j_y^2 \\ p_{xy}^{(\mathrm{eq})} &= j_x j_y. \end{aligned} \tag{9}$$

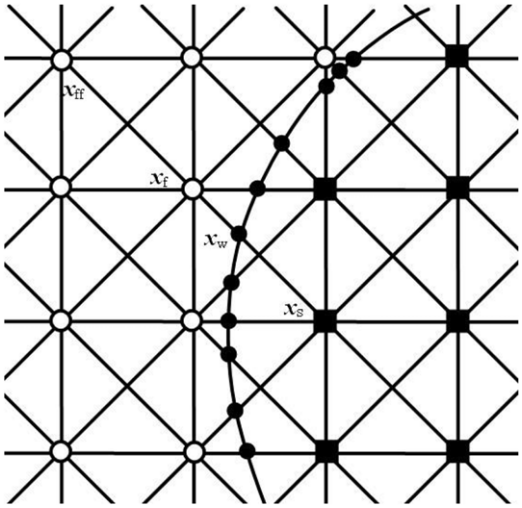


Fig. 2. Illustration of curved boundary (the empty circles (◯) represent the fluid nodes, the solid squares (■) represent the solid nodes, and the solid circles (•) represent the boundary nodes).

The evolution progress includes two steps, collision and advection, which can be respectively described by

$$|f^+(\boldsymbol{x}, t)\rangle = |f(\boldsymbol{x}, t)\rangle - \boldsymbol{M}^{-1}\boldsymbol{S}[|m(\boldsymbol{x}, t)\rangle - |m^{(\mathrm{eq})}(\boldsymbol{x}, t)\rangle], \tag{10a}$$

and

$$|f(\boldsymbol{x} + \boldsymbol{e}\delta t, t + \delta t)\rangle = |f^+(\boldsymbol{x}, t)\rangle. \tag{10b}$$

The macroscopic density and velocity can be respectively calculated using

$$\rho = \sum_{i=0}^{8} f_i, \tag{11a}$$

and

$$\boldsymbol{u} = \frac{1}{\rho}\sum_{i=0}^{8} f_i \boldsymbol{e}_i. \tag{11b}$$

The pressure p can be obtained by

$$p = \rho c_s^2. \tag{12}$$

2.2. Boundary treatment

The unified boundary treatment scheme is a bounce-back interpolation method with second order accuracy which was proposed by Yu et al. in [8] and is also referred to as YMS scheme. This scheme uses a variable defined as

$$q = (|\boldsymbol{x}_f - \boldsymbol{x}_w|)/(|\boldsymbol{x}_f - \boldsymbol{x}_s|) \tag{13}$$

for interpolation purpose, where $\boldsymbol{x}_w$ is a boundary node, $\boldsymbol{x}_s$ is a solid node, and $\boldsymbol{x}_f$ is a fluid node. Fig. 2 illustrates an example of the YMS scheme. At time t, the YMS scheme assumes that, after collision, the distribution function can pass through the boundary node, $\boldsymbol{x}_w$, and reach the solid node, $\boldsymbol{x}_s$, such that

$$f_i(\boldsymbol{x}_s, t + \delta t) = f_i^+(\boldsymbol{x}_f, t). \tag{14}$$

Thus, the distribution function can be obtained through the following interpolation:

$$f_i(\boldsymbol{x}_w, t + \delta t) = (1 - q)f_i(\boldsymbol{x}_f, t + \delta t) + qf_i(\boldsymbol{x}_s, t + \delta t). \tag{15}$$

The following assumption has to be adopted to satisfy the no-slip boundary condition for solid wall:

$$f_{\bar{i}}(\boldsymbol{x}_w, t + \delta t) = f_i(\boldsymbol{x}_w, t + \delta t), \tag{16}$$

where the subscript $\bar{i}$ represents the opposite of the ith discrete velocity direction, that is, $\bar{i} = -i$. Finally, the second order interpolation that involves the fluid node $\boldsymbol{x}_{ff}$ and boundary node $\boldsymbol{x}_w$, as illustrated in Fig. 2, applies, leading to the following expression of the expected distribution function at the fluid node, $\boldsymbol{x}_f$:

$$\begin{aligned} f_{\bar{i}}(\boldsymbol{x}_f, t+\delta t) &= \frac{1}{1+q}(q f_{\bar{i}}(\boldsymbol{x}_{ff}, t+\delta t) + f_{\bar{i}}(\boldsymbol{x}_w, t+\delta t)) \\ &= \frac{1}{1+q}(q f_{\bar{i}}^{+}(\boldsymbol{x}_f, t) + q f_i^{+}(\boldsymbol{x}_f, t) + (1-q) f_i^{+}(\boldsymbol{x}_{ff}, t)). \end{aligned} \tag{17}$$

2.3. Turbulence model

In order to solve the turbulent flow problems, the Spalart–Allmaras (SA) turbulence model [26] is employed. The SA turbulence model features employing a single equation describing the transformed eddy kinematic viscosity $\tilde{\nu}$:

$$\frac{D\tilde{\nu}}{Dt} = C_{b_1}(1-f_{t_2})\tilde{S}\tilde{\nu} + \frac{1}{\sigma}[\nabla((\nu_0+\tilde{\nu})\nabla\tilde{\nu}) + C_{b_2}(\nabla\tilde{\nu})^2] - \left(C_{w_1} f_w - \frac{C_{b_1}}{\kappa^2} f_{t_2}\right)\left(\frac{\tilde{\nu}}{d}\right)^2, \tag{18}$$

where $\tilde{\nu}$ can be related to the original eddy kinematic viscosity ν_t via

$$\nu_t = \tilde{\nu} f_{v_1} \tag{19}$$

d is the distance between the fluid node and the wall boundary, and

$$\begin{aligned} &f_{t_2} = C_{t_3}\exp(-C_{t_4}\chi^2), \\ &\tilde{S} = \sqrt{2\Omega_{ij}\Omega_{ij}} f_{v_3} + \frac{\tilde{\nu}}{\kappa^2 d^2} f_{v_2}, \qquad \Omega_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right), \\ &f_{v_2} = \left(1+\frac{\chi}{C_{v_2}}\right)^{-3}, \qquad f_{v_3} = \frac{(1+\chi f_{v_1})(1-f_{v_2})}{\chi}, \qquad C_{v_2} = 5, \qquad \chi = \max(\chi, 10^{-4}), \\ &f_w = g\left(\frac{1+{C_{w_3}}^6}{g^6+{C_{w_3}}^6}\right)^{\frac{1}{6}}, \qquad g = r + C_{w_2}(r^6 - r), \qquad r = \frac{\tilde{\nu}}{\tilde{S}\kappa^2 d^2}. \end{aligned} \tag{20}$$

with the constants used in this study being $C_{b_1} = 0.1355$, $C_{b_2} = 0.622$, $\sigma = 2/3$, $\kappa = 0.41$, $C_{v_1} = 7.1$, $C_{t_3} = 1.1$, $C_{t_4} = 2$, $C_{w_1} = \frac{C_{b_1}}{\kappa^2} + \frac{1+C_{b_2}}{\sigma}$, $C_{w_2} = 0.3$, and $C_{w_3} = 2$. The other variable on the right hand side of Eq. (19) is defined as

$$f_{v_1} = \frac{\chi^3}{\chi^3 + {C_{v_1}}^3}, \tag{21}$$

where

$$\chi = \frac{\tilde{\nu}}{\nu_0} \tag{22}$$

with ν_0 representing the laminar kinematic viscosity. More details about the above coefficients chosen for the SA model can be found in [27,28].

After adding the SA model, the kinematic viscosity is modified as

$$\nu = \nu_0 + \nu_t. \tag{23}$$

Then, the local relaxation time τ can be updated by using

$$\tau = \frac{3\nu}{\delta t} + \frac{1}{2}. \tag{24}$$

Note that [26–28] provide detailed discretization schemes and boundary condition prescription for solving Eq. (18).

2.4. Advection on non-uniform mesh

On a uniform mesh, advection can be realized directly by Eq. (10b), which signifies the distribution function can advect from one node to its neighboring node within one time step. However, when a non-uniform mesh is employed, the time step, δt, is chosen in a conservative way such that its suitability can be guaranteed for the locally finest mesh, e.g., the cell formed by $\boldsymbol{x}$, $\boldsymbol{x}_1$, $\boldsymbol{x}_4$, and $\boldsymbol{x}_3$ as shown in the left part of Fig. 3. However, for a neighboring cell with bigger area (due to non-uniformity of the mesh), e.g., the cell formed by $\boldsymbol{x}_4$, $\boldsymbol{x}_5$, $\boldsymbol{x}_8$, and $\boldsymbol{x}_7$, the advection over the conservatively selected time step, δt, is very unlikely to cover the entire diagonal of the bigger cell. To accommodate the non-uniformity of the mesh that has to be applied to sensitive regions such as the solid object involved in the flow problem along with the vicinity of the object,

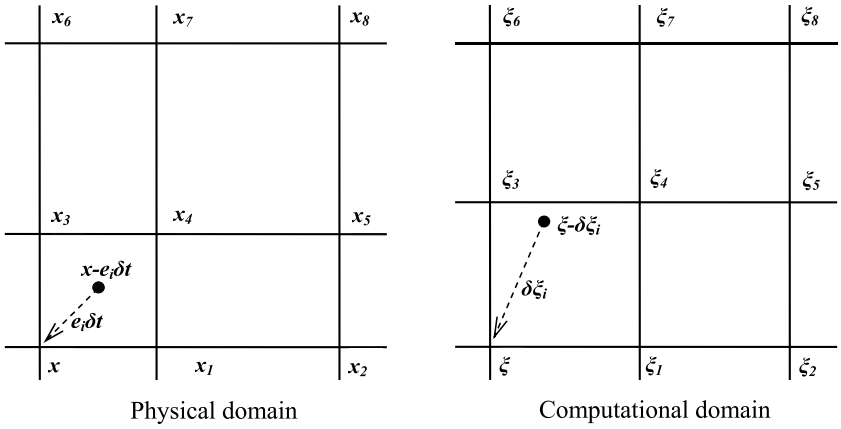


Fig. 3. Advection on a non-uniform mesh.

the generalized form of interpolation supplemented lattice Boltzmann method [11,29] has to be adopted. Using the example as shown in Fig. 3, on the computational domain the advection term is computed using

$$f_i(\xi, t+\delta t) = f_i^{+}(\xi - \delta\xi_i, t), \tag{25}$$

with

$$\delta\xi_i = \int_0^{\delta t} \tilde{\boldsymbol{e}}_i dt, \tag{26}$$

where ξ is the coordinate in computational domain, $\tilde{\boldsymbol{e}}_i$ is the contravariant velocity of the discrete velocity, $\boldsymbol{e}_i$, satisfying

$$\tilde{e}_{i\alpha} = e_{i\beta}\frac{\partial \xi_\alpha}{\partial x_\beta} \tag{27}$$

where the summation convention is used with the subscripts α and β. To improve the numerical solution accuracy, the right hand side of Eq. (25) is calculated using the second-order upwind interpolation scheme in its two-dimensional form.

2.5. Numerical procedure

The numerical procedure over one time step is implemented as follows:

1. Compute moments using Eq. (6) from given distribution function;
2. Execute the collision in momentum space $|m^{+}\rangle = |m\rangle - \boldsymbol{S}[|m\rangle - |m^{(\mathrm{eq})}\rangle]$;
3. Transform moments to distribution functions $|f\rangle = \boldsymbol{M}^{-1}|m\rangle$;
4. Execute the advection;
5. Use the YMS scheme for curved boundary treatment;
6. Compute macroscopic density and velocity;
7. Modify the kinematic viscosity and update relaxation time;
8. Exit if satisfactory convergence is achieved; otherwise, go back to Step 1.

3. Numerical simulations

Selected test cases corresponding to turbulent flows around an NACA0012 airfoil and a flat plate are presented here, and comparison is made against other available numerical results reported in [23] and [24], respectively. The SA turbulence model is employed in this present study. Throughout this section, the fluid density is taken as $\rho = 1.0$, and the free stream velocity is $U_\infty = 0.1$ with D2Q9 lattice sound speed referenced.

The airfoil and plate lengths are both chosen as characteristic length and viewed as 1. The computational domain, with the obstacle practically placed in the center, occupies a 40×40 square to ensure the appropriateness of imposing the unperturbed free stream velocity at the inlet as well as the upper and lower boundaries of the domain; at the outlet, a convective boundary condition [24] is applied so that vortices in the wake region are allowed to naturally cross the outflow boundary while the flow behavior within the computational domain can stay unaffected. The grid granularity in the central obstructed zone of the domain is uniform and fairly fine; beyond that zone, the grid size grows gradually towards the outer boundaries.

All test cases reported in this paper were run on an NVIDIA Tesla C1060 Graphic Processing Unit (GPU) via the Compute Unified Device Architecture (CUDA) interface [30,31]. By utilizing the explicit parallelism provided by the graphics hardware, computing time can be greatly saved. Compared to the time consumption by using a CPU (Intel(R) Xeon(R) CPU E5520 @ 2.27 GHz), test cases of this present study employing GPU may generally reduce the computational time by 90%.

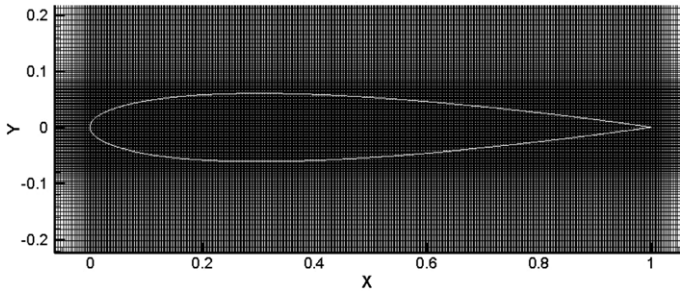


Fig. 4. Mesh zoomed into the vicinity of airfoil.

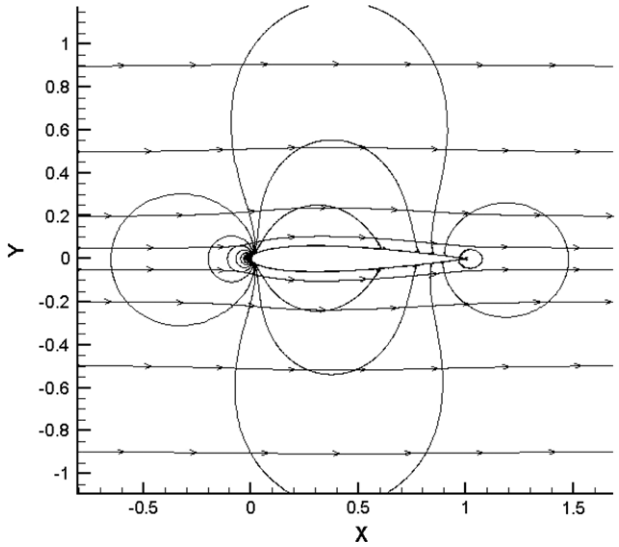


Fig. 5. Streamline and pressure contours in the vicinity of the airfoil ($\mathrm{Re} = 5 \times 10^5$, $\alpha = 0^\circ$).

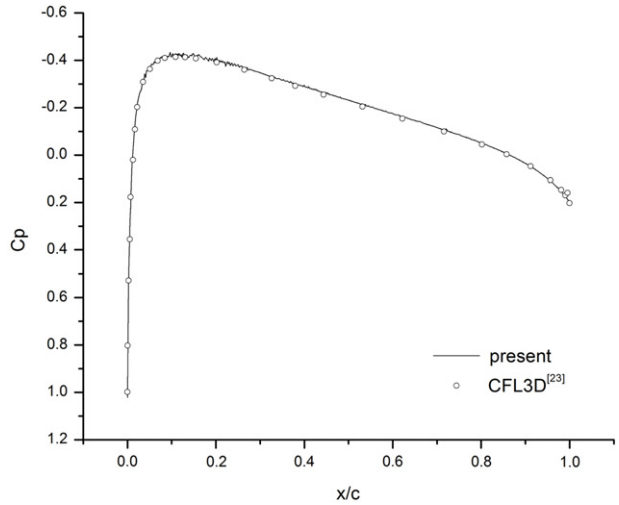


Fig. 6. Surface pressure coefficients ($\mathrm{Re} = 5 \times 10^5$, $\alpha = 0^\circ$).

3.1. Unseparated flow around an NACA0012 airfoil

The turbulent flow at $\mathrm{Re} = 5 \times 10^5$ around an NACA0012 airfoil is investigated using a 5209×813 non-body-fitted Cartesian mesh; Fig. 4 focuses on the obstructed area of the mesh, in which the local grid granularity is as small as 0.0002. Employing this mesh, flows of incidence-free ($\alpha = 0^\circ$) and slight incidence ($\alpha = 7^\circ$) are both simulated and compared

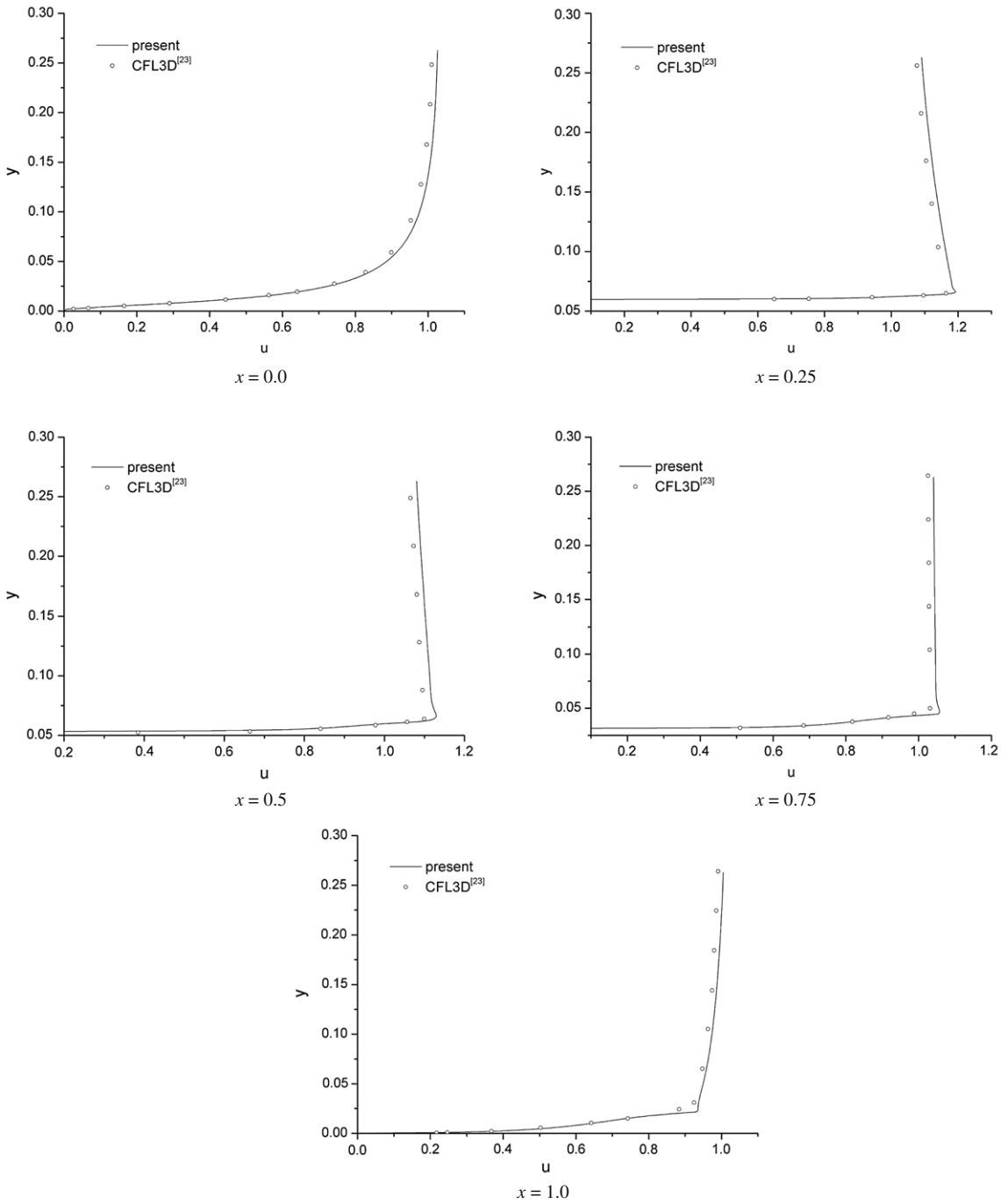


Fig. 7. The x-component of velocity at $x/c = 0.0, 0.25, 0.5, 0.75$, and 1.0 ($\mathrm{Re} = 5 \times 10^5$, $\alpha = 0^\circ$).

with the CFL3D simulation results [23] that were obtained using finite volume discretization of the Reynolds-averaged Navier–Stokes equations. A priori, since the SA turbulent flow model has been employed here and the incidence is either zero or very small, no flow separation is expected in the simulation results though the Reynolds number has reached 5×10^5.

3.1.1. Flow with nil incidence ($\alpha = 0^\circ$)

The first numerical experiment chooses the incidence-free inflow condition, which helps testify the accuracy of the code in terms of preservation of solution symmetry. Fig. 5 groups both streamline and pressure contours in the vicinity of the airfoil. Evidently, for this zero-incidence flow, both results appear strictly symmetric about the chord of the horizontally placed symmetric airfoil; besides, no flow separation is indicated by the streamline or pressure behavior.

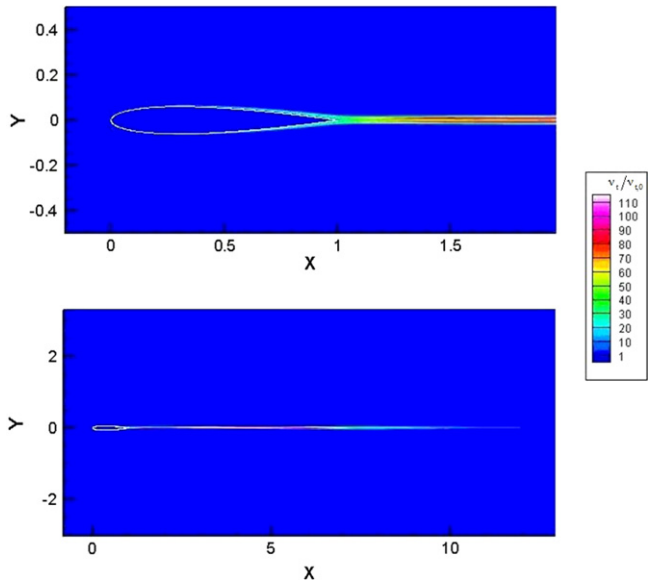


Fig. 8. Viscosity ratio contours ($\mathrm{Re} = 5 \times 10^5$, $\alpha = 0^\circ$).

For a more careful examination, Fig. 6 depicts the pressure coefficient distribution only on the upper surface of the airfoil since the solution symmetry has been verified in this case. The pressure coefficient obtained in this study looks in very good agreement with the CFL3D simulation result as reported in [23]. Some slight fluctuation is found in the neighborhood of the minimum C_p value. This suggests the necessity of employing a higher order upwind interpolation than the second order scheme implemented in this numerical experiment. In the portion near the trailing edge, which is another region of strong sensitivity, a fairly smooth C_p distribution is achieved that obviously outperforms the CFL3D, as indicated by the comparison shown in Fig. 6.

Fig. 7 illustrates the profiles of the x-component of velocity at five equally spaced locations on the upper surface of the airfoil, along with a complete series of detailed comparisons made against their respective CFL3D results. For all sample locations, the two sets of results agree with each other in the overall trend and, moreover, on quantitative basis.

In order to examine the turbulence characteristics of this high Reynolds number flow, the contours of the ratio of the eddy kinematic viscosity ν_t to the initial eddy kinematic viscosity $\nu_{t,0}$, which will be subsequently referred to as the viscosity ratio, are both locally (closer to the obstacle) and more globally (focusing more on the wake region) shown in Fig. 8. The symmetry of the solution is here once more observed. The flow turbulence mainly appears in the wake region with the maximum viscosity ratio towards 110. Since the computation domain is built with a larger dimension behind the trailing edge, at the exit the outflow looks already uniform.

3.1.2. *Slightly inclined flow* ($\alpha = 7^\circ$)

Now, the free stream is a little inclined for a new set of numerical experiments. The angle of attack at $\alpha = 7^\circ$, which was used in the CFL3D simulation [23], is chosen here so that a systematic comparison is feasible. Similar to the figure plotting arrangements made in the precedent test case, the streamline and pressure contours are shown together in Fig. 9. In spite of a high Reynolds number and a small angle of attack, the streamline pattern in Fig. 9 demonstrates no indication of flow separation owing to the incorporation of turbulence model into computation. Due to incidence of the incoming flow, solutions are no longer symmetric.

Fig. 10 shows that the pressure coefficients on the upper and lower surfaces of the airfoil are obviously not symmetric, either. On both surfaces, the pressure result appears in good agreement with the CFL3D data. The overall effect elucidated by the pressure solution would be lifting as a result of the incidence of the incoming flow.

Then, the x-components of the velocity at five positions on both lower and upper surfaces of the airfoil are plotted in Fig. 11. Again, all the shown profiles obtained in this study have nearly the same shape and magnitude with the CFL3D results. Compared with the lower surface, a greater x-component of the velocity on the upper surface is noticed for all the five sample positions. Among the five pairs of comparisons illustrated in Fig. 11, the velocity profiles on the upper and lower surfaces differ most remarkably at the leading edge of the airfoil, which echoes Fig. 10 that exhibits the most significant dissymmetry of the surface pressure coefficient near the leading edge as well.

The viscosity ratio is plotted in Fig. 12 in the same manner as used in Fig. 8 for the case of nil incidence flow. Unlike the precedent case, now the contour is no longer symmetric, which is again due to the presence of an angle of attack of the inflow. The maximum ratio is about 100, which is close to that in the previous case; this suggests the angle of attack $\alpha = 7^\circ$

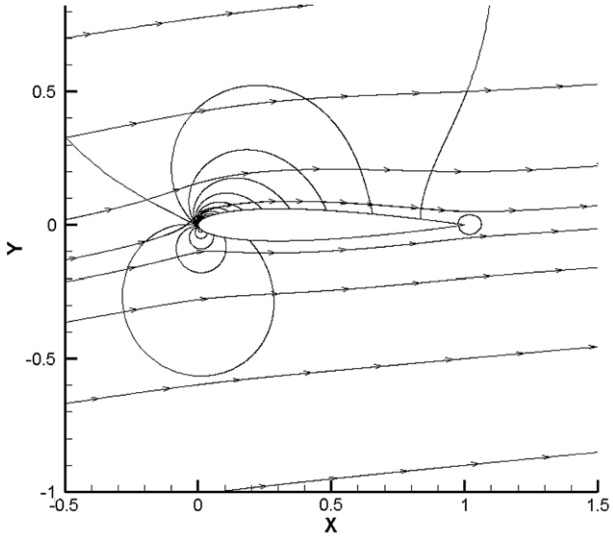


Fig. 9. Streamline and pressure contours in the vicinity of the airfoil ($\mathrm{Re} = 5 \times 10^5$, $\alpha = 7°$).

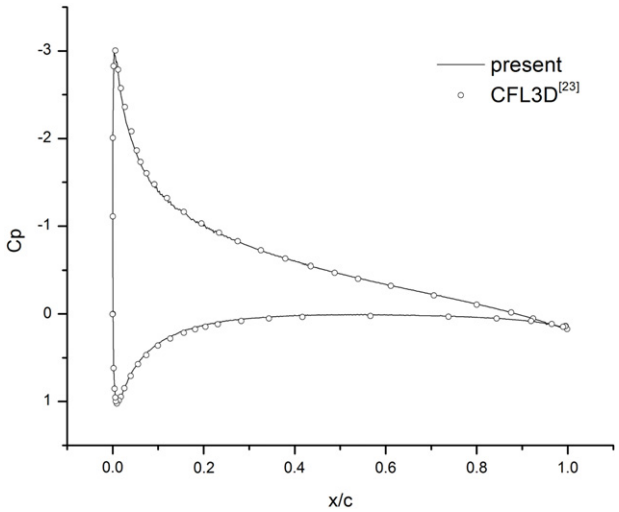


Fig. 10. Surface pressure coefficients ($\mathrm{Re} = 5 \times 10^5$, $\alpha = 7°$).

is not large enough to significantly alert the effect of turbulence. The uniformity of this ratio at the exiting boundary signifies once more the satisfaction of the dimensions chosen for this simulation.

3.2. Separated turbulent flow around a flat plate

This case uses a flat plate of uniform finite thickness (0.1% of the plate length) as obstacle and assigns a relatively large angle of attack ($\alpha = 18°$) along with $\mathrm{Re} = 2 \times 10^4$ for the incoming flow, which make some back flow phenomena expectable in the computation results. The non-body-fitted Cartesian mesh resolution is 2209×829. Fig. 13 provides an enlarged view of the mesh in the vicinity of the obstacle where the minimum local grid granularity reaches 0.0005. The reference results for this case are extracted from [24] in which the Reynolds number averaged Navier–Stokes (RANS) and large eddy simulation (LES) approaches were both employed.

The contours of averaged pressure and streamline results are plotted in Figs. 14 and 15, respectively. Both indicate a strong flow recirculation above the upper surface of the plate, which represents the primary eddy in this flow scenario. Through a detailed comparison with [24], Fig. 16 reveals that the present approach and the RANS, LES approaches may result in a nearly identical overall trend of the pressure coefficient on both surfaces of the plate. On the upper surface that is beneath the aforementioned large eddy, the insignificant variation of the pressure suggests the presence of a stalled configuration. In terms of the magnitude of the pressure coefficient on the upper surface, the present result is slightly lower

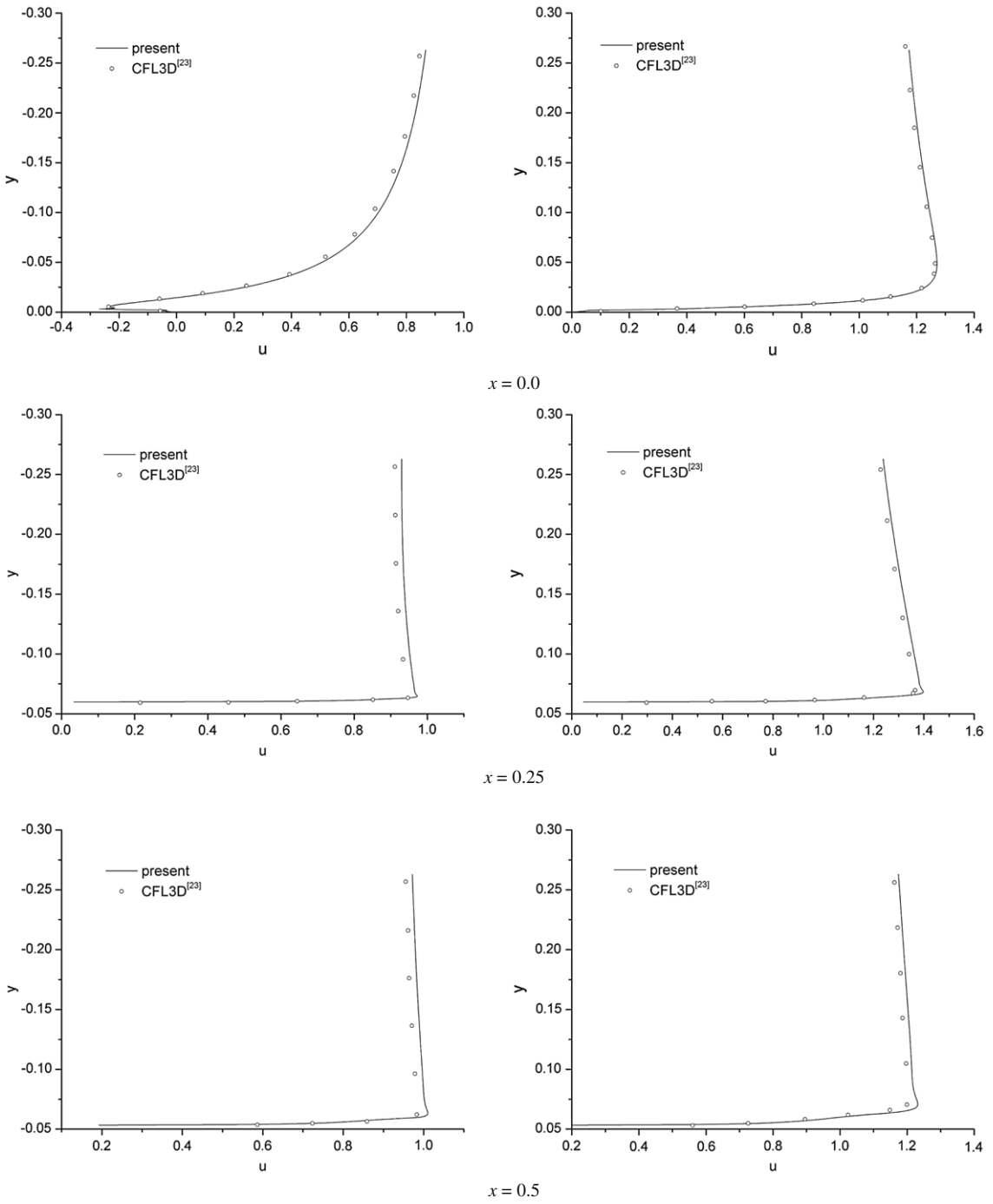


Fig. 11. The x-component of velocity at $x/c = 0.0, 0.25, 0.5, 0.75$, and 1.0 ($\mathrm{Re} = 5 \times 10^5$, $\alpha = 7°$, left: lower surface, right: upper surface).

than those obtained using RANS and LES [24]; this consequently results in smaller drag and lift coefficients as indicated by the comparison shown in Table 1.

A more thorough interpretation of the conducted simulation can be accomplished by examining the velocity profiles as shown in Fig. 17. The tangential velocity components of the averaged flow field are displayed at three different locations above the plate ($x/c = 0.2$, 0.6 and 1.0), followed by a location in the wake ($x/c = 1.2$). Once again, at each of the four sample locations, the velocity profile has the same overall trend with the RANS and LES results [24]. The negative tangential velocity component clearly reveals the presence of vortex above the plate.

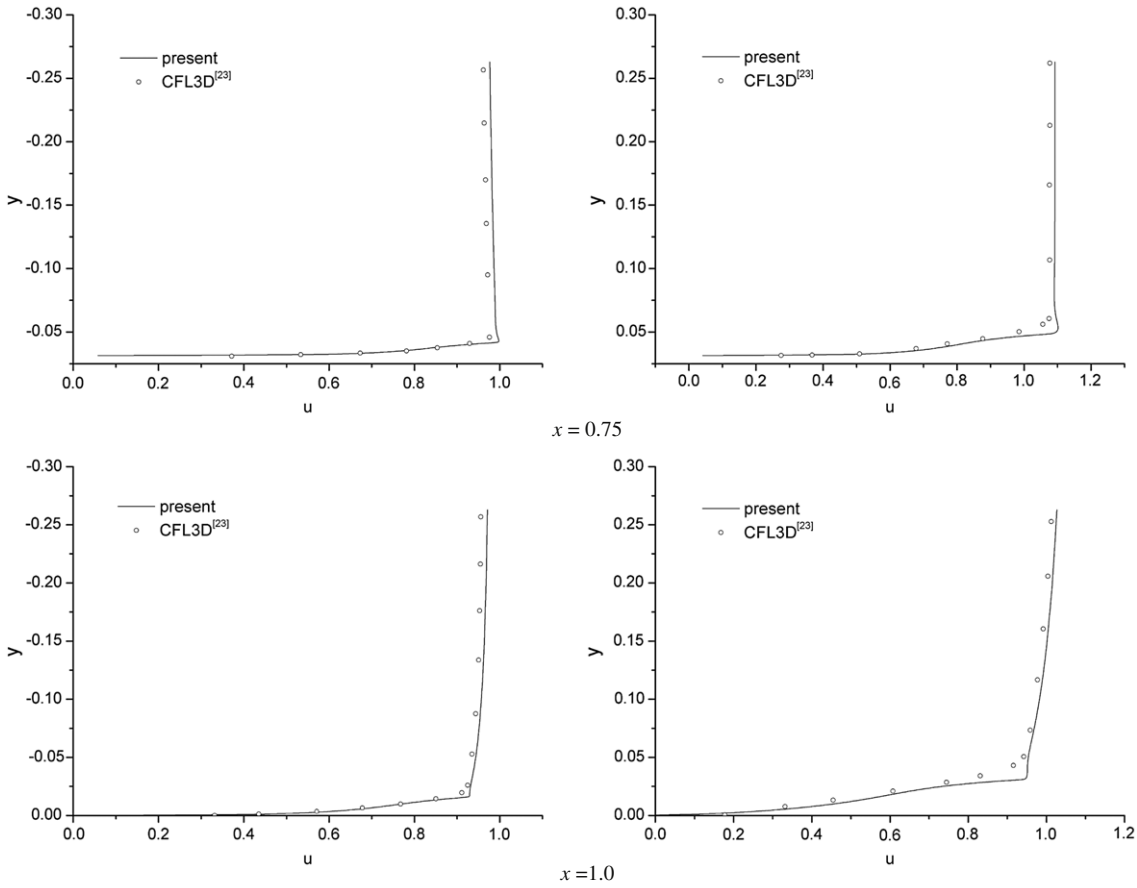


Fig. 11. (*continued*)

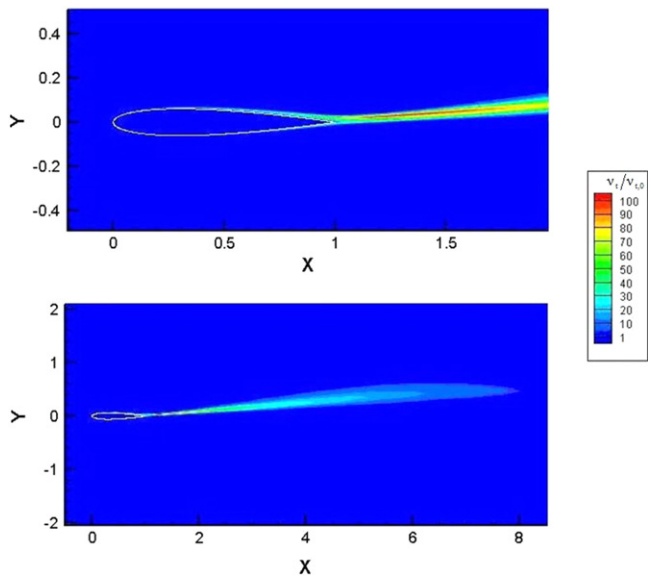


Fig. 12. Viscosity ratio contours ($\mathrm{Re} = 5 \times 10^5$, $\alpha = 7°$).

Finally, Fig. 18 illustrates the contour of the viscosity ratio in a fashion similar to the previous airfoil cases. Unlike those unseparated flows around the airfoil, here new flow characteristics can be noticed from the viscosity ratio contour. First, the magnitude of the ratio now rises to 220 at the center of the primary vortex, while the airfoil cases record the ratio up to 110 only. Second, the ratio contour looks obviously weaving in the wake region whereas the airfoil cases generally yield a much

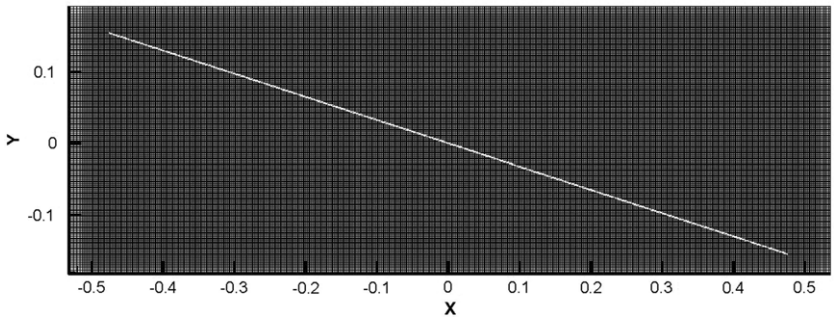


Fig. 13. Mesh zoomed into the vicinity of flat plate.

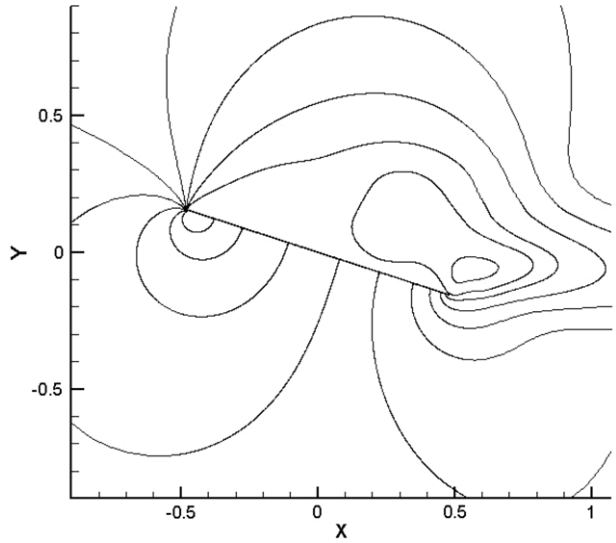


Fig. 14. Pressure contours around the flat plate.

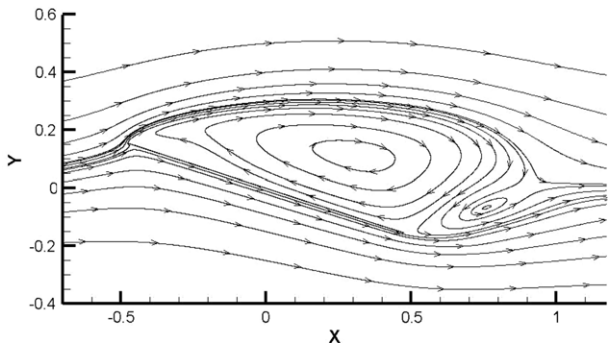


Fig. 15. Streamline contours around the flat plate.

Table 1
Comparison of drag and lift coefficients.

	$\bar{c}_l$	$\bar{c}_d$
RANS	1.318	0.439
LES	1.128	0.380
Present	1.090	0.367

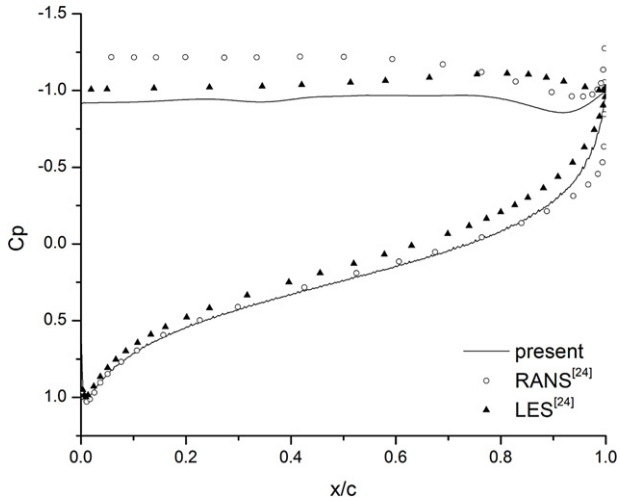


Fig. 16. Plate surface pressure coefficient distribution.

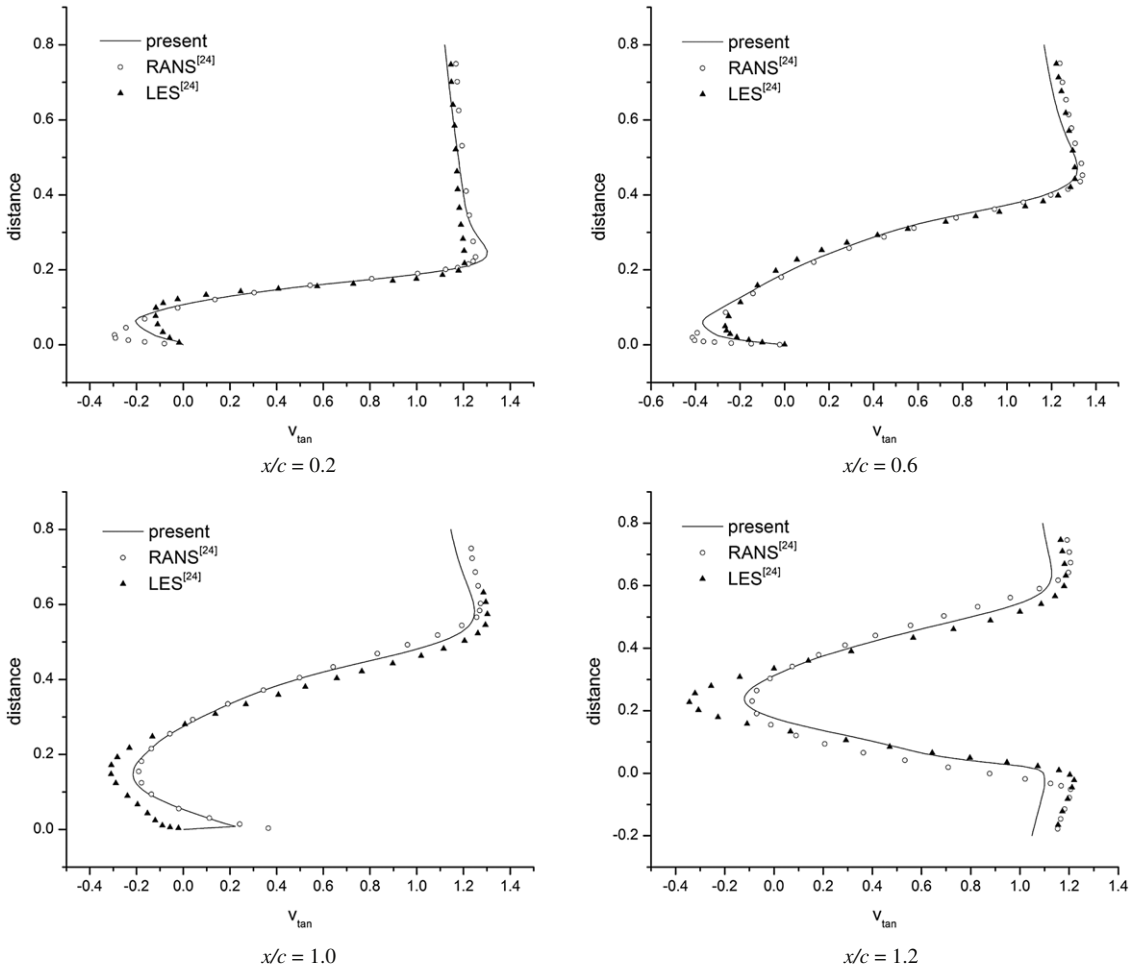


Fig. 17. Profiles of the tangential velocity vs. wall-normal distance at four different locations of the flat plate.

more regular trace for the ratio in that area. The presence of such more evident turbulent flow effects, when compared with the previous NACA airfoil cases, is physically attributed to a largely increased incidence of this flow passing through the flat plate.

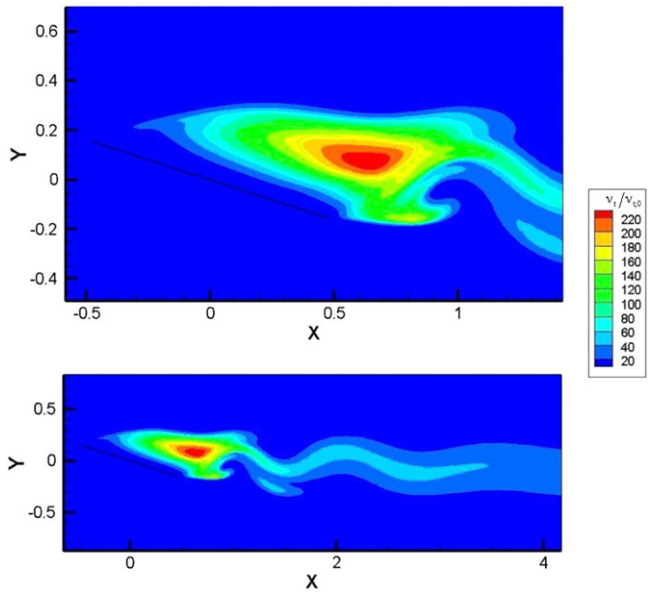


Fig. 18. Viscosity ratio contours around the flat plate.

4. Conclusion

In this paper, the YMS unified boundary treatment scheme is introduced to the multi-relaxation-time (MRT) lattice Boltzmann method (LBM) to make a Cartesian mesh usable in flow simulations involving configurations of curved boundary. The Spalart–Allmaras (SA) turbulence model is also employed to describe the turbulent flow effects. Numerical experiments include a flow at $\mathrm{Re} = 5 \times 10^5$ around an NACA0012 airfoil for both nil incidence and non-nil incidence cases, and a flow at $\mathrm{Re} = 2 \times 10^4$ around a largely inclined flat plate. All numerical results obtained from this present approach satisfactorily agree with other simulation data reported in the available literature. Therefore, the LBM with enriched ingredients proposed in this study is capable for simulation of highly turbulent viscous flows. Owing to the simplicity of Cartesian mesh generation and the appealing numerical properties of MRT, it looks promising for more extensive applications of this enhanced LBM approach.

Acknowledgments

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